Multisensory Fitness, Inc.

EMC TEST REPORT FOR

SMARTfit ProTrainer 3 CE-Black
Models: 1-20222 & 1-20102

Tested to The Following Standards:

EN 55032 (2012) +C1, EN 61000-3-2 (2014),
EN 61000-3-3 (2013) and EN 55024 (2010) +A1

Report No.: 97844-4

Date of issue: July 15, 2016

A2LA ACCREDITED
Testing Certificates:
803.01, 803.02, 803.05, 803.06

This test report bears the accreditation symbol indicating that the testing performed herein meets the test and reporting requirements of ISO/IEC 17025 under the applicable scope of EMC testing for CKC Laboratories, Inc.

We strive to create long-term, trust based relationships by providing sound, adaptive, customer first testing services. We embrace each of our customers' unique EMC challenges, not as an interruption to set processes, but rather as the reason we are in business.

Testing the Future
CKC LABORATORIES, INC.

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ADMINISTRATIVE INFORMATION

Test Report Information

REPORT PREPARED FOR:

Multisensory Fitness, Inc.
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Oxnard, CA 93033

Representative: Jim Manley
Customer Reference Number: 00002104

REPORT PREPARED BY:

Dianne Dudley
CKC Laboratories, Inc.
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Mariposa, CA 95338

Project Number: 97844

DATE OF EQUIPMENT RECEIPT: July 8, 2016
DATE(S) OF TESTING: July 8-11, 2016

Report Authorization

The test data contained in this report documents the observed testing parameters pertaining to and are relevant for only the sample equipment tested in the agreed upon operational mode(s) and configuration(s) as identified herein. Compliance assessment remains the client's responsibility. This report may not be used to claim product endorsement by A2LA or any government agencies. This test report has been authorized for release under quality control from CKC Laboratories, Inc.

Steve Behm
Director of Quality Assurance & Engineering Services
CKC Laboratories, Inc.

Test Facility Information



Our laboratories are configured to effectively test a wide variety of product types. CKC utilizes first class test equipment, anechoic chambers, data acquisition and information services to create accurate, repeatable and affordable test results.

TEST LOCATION(S):
CKC Laboratories, Inc.
110 Olinda Place
Brea, CA 92823

Software Versions

CKC Laboratories Proprietary Software	Version
EMITest Emissions	5.03.02
EMITest Immunity	5.03.02

Site Registration & Accreditation Information

Location	CB #	TAIWAN	CANADA	FCC	JAPAN
Brea D	US0060	SL2-IN-E-1146R	3082D-2	100638	A-0147

SUMMARY OF RESULTS

Standard / Specification: EN 55032 (2012) +C1

Test Procedure	Description	Modifications	Results
EN 55032 Class A	Conducted Emissions	Mod. #1	Pass
EN 55032 Class A	Telecom Conducted Emissions	NA	NA
EN 55032 Class A	Radiated Emissions	Mod. #1	Pass

NA = Not Applicable

Standard / Specification: EN 61000-3-2 (2014)

Test Procedure	Description	Modifications	Results
EN 61000-3-2	Harmonic Emissions	Mod. #1	Pass

Standard / Specification: EN 61000-3-3 (2013)

Test Procedure	Description	Modifications	Results
EN 61000-3-3	Voltage Fluctuations and Flicker Emissions	Mod. #1	Pass

Standard / Specification: EN 55024 (2010) +A1

Test Procedure	Description	Modifications	Results
EN 61000-4-2 (2009)	Electrostatic Discharge	Mod. #1	Pass
EN 61000-4-3 (2006)+A1+A2	Radiated Immunity	Mod. #1	Pass
EN 61000-4-4 (2004)	Electrical Fast Transient Burst	Mod. #1	Pass
EN 61000-4-5 (2006)	Surge	Mod. #1	Pass
EN 61000-4-6 (2009)	Conducted Immunity	Mod. #1	Pass
EN 61000-4-8 (2010)	Magnetic Immunity	NA	NA
EN 61000-4-11 (2004)	Voltage Dips & Interrupts	Mod. #1	Pass

NA = Not Applicable

Modifications During Testing

This list is a summary of the modifications made to the equipment during testing.

Summary of Conditions
Mod #1: Grounded the shield of the three panel cables internal to the controller by using a single wire from the power supply ground terminal. Grounded the dc minus using the same single wire from the power supply ground terminal. Removed the approximately 100mm length extension cable from the top six LED boards.

Modifications listed above must be incorporated into all production units.

Conditions During Testing

This list is a summary of the conditions noted to the equipment during testing.

Summary of Conditions
None

EQUIPMENT UNDER TEST

During testing numerous configurations may have been utilized. The configurations listed below support compliance to the standard(s) listed in the Summary of Results section.

Configuration 2

Equipment Tested:

Device	Manufacturer	Model #	S/N
SMARTfit ProTrainer 3 CE-Black	Multisensory Fitness, Inc.	1-20222	None

Support Equipment:

Device	Manufacturer	Model #	S/N
None			

Configuration 3 = Configuration A

Equipment Tested:

Device	Manufacturer	Model #	S/N
SMARTfit ProTrainer 3 CE-Black	Multisensory Fitness, Inc.	1-20102	None

Support Equipment:

Device	Manufacturer	Model #	S/N
None			

EN 55032 EMISSIONS

Conducted Emissions

Test Notes: Conducted Disturbances at Mains Terminals, LISN method.

Test Data

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**EN 55032 AC Mains Class A - Average**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Conducted Emissions**	Time:	3:52:49 PM
Tested by:	S. Yamamoto	Sequence#:	5
Software:	EMITest 5.03.02		230V 50Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 2			

Support Equipment:

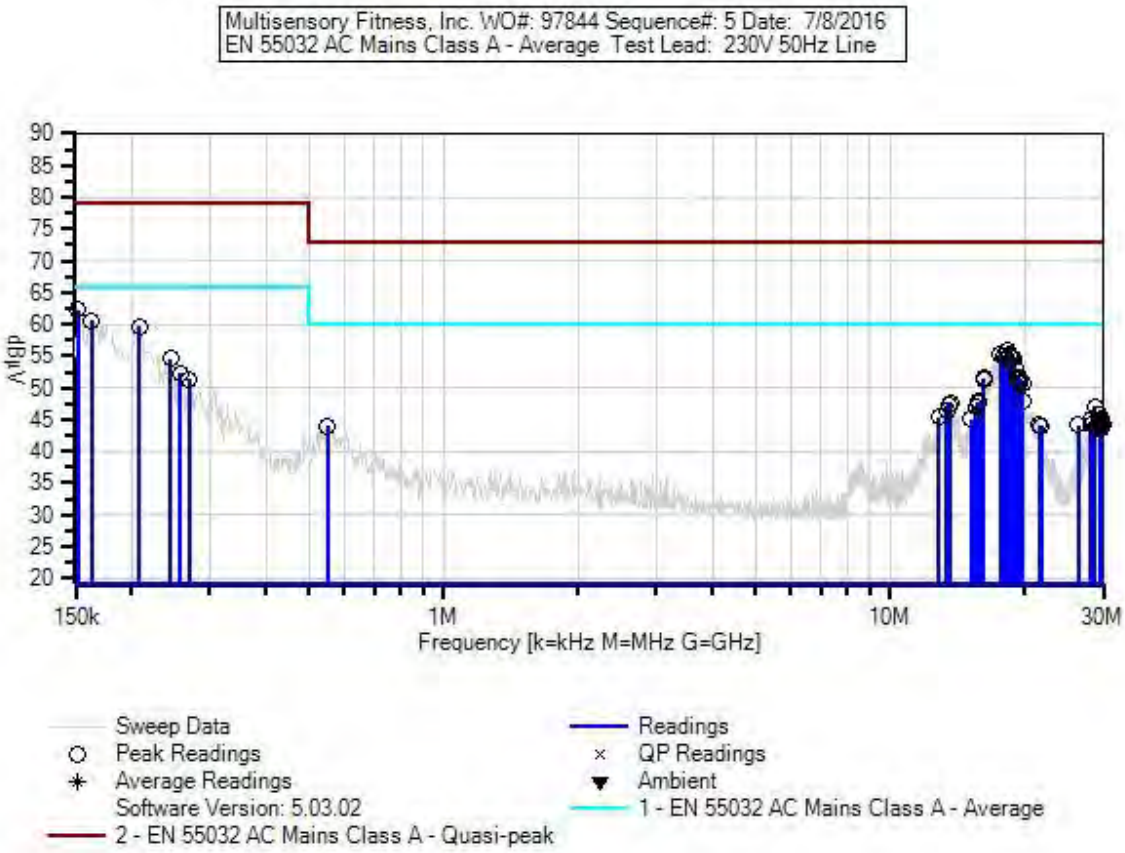
Device	Manufacturer	Model #	S/N
Configuration 2			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 150kHz to 30MHz. RBW=VBW=9kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
Modification 1 was in place during testing.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 5 Date: 7/8/2016
EN 55032 AC Mains Class A - Average Test Lead: 230V 50Hz Line
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
dBµV
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Peak Readings
Average Readings
Software Version: 5.03.02
2 - EN 55032 AC Mains Class A - Quasi-peak
Readings
QP Readings
Ambient
1 - EN 55032 AC Mains Class A - Average

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T1	AN02343	High Pass Filter	HE9615-150K-50-720B	1/8/2015	1/8/2017
T2	ANP01910	Cable	RG-142	11/30/2015	11/30/2017
T3	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T4	ANP06085	Attenuator	SA18N10W-09	12/17/2014	12/17/2016
T5	AN00969A	50uH LISN-Line 1 (L1) (dB)	3816/2NM	3/12/2015	3/12/2017
	AN00969A	50uH LISN-Line 2 (L2) (dB)	3816/2NM	3/12/2015	3/12/2017

Measurement Data: Reading listed by margin. Test Lead: Line

#	Freq	Rdng	T1 T5	T2	T3	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV	dBµV	dB	Ant
1	151.818k	53.5	+3.0 +0.1	+0.0	+0.0	+5.7	+0.0	62.3	66.0	-3.7	Line
2	18.381M	49.0	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	55.9	60.0	-4.1	Line
3	17.614M	48.5	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	55.3	60.0	-4.7	Line
4	18.489M	48.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	55.3	60.0	-4.7	Line
5	18.029M	48.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	55.1	60.0	-4.9	Line
6	18.534M	48.1	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	55.0	60.0	-5.0	Line
7	18.164M	47.9	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	54.8	60.0	-5.2	Line
8	18.832M	47.9	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	54.8	60.0	-5.2	Line
9	162.726k	54.4	+0.4 +0.1	+0.0	+0.0	+5.7	+0.0	60.6	66.0	-5.4	Line
10	18.670M	47.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	54.5	60.0	-5.5	Line
11	208.176k	53.8	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	59.7	66.0	-6.3	Line
12	19.012M	46.3	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	53.2	60.0	-6.8	Line
13	19.346M	44.7	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	51.6	60.0	-8.4	Line
14	16.233M	44.7	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	51.5	60.0	-8.5	Line
15	19.436M	44.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	51.5	60.0	-8.5	Line
16	19.464M	44.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	51.5	60.0	-8.5	Line
17	19.563M	44.4	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	51.4	60.0	-8.6	Line

18	16.215M	44.4	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	51.2	60.0	-8.8	Line
19	19.500M	43.8	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	50.8	60.0	-9.2	Line
20	19.797M	43.8	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	50.8	60.0	-9.2	Line
21	244.536k	48.6	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	54.5	66.0	-11.5	Line
22	19.815M	41.0	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	48.0	60.0	-12.0	Line
23	15.809M	41.0	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	47.8	60.0	-12.2	Line
24	13.635M	41.1	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.6	60.0	-12.4	Line
25	15.737M	40.8	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	47.6	60.0	-12.4	Line
26	13.527M	40.8	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.3	60.0	-12.7	Line
27	28.703M	39.7	+0.2 +0.8	+0.2	+0.5	+5.7	+0.0	47.1	60.0	-12.9	Line
28	15.584M	40.1	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	46.9	60.0	-13.1	Line
29	13.373M	39.8	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.3	60.0	-13.7	Line
30	257.262k	46.3	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	52.2	66.0	-13.8	Line
31	29.533M	38.3	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.8	60.0	-14.2	Line
32	12.868M	38.9	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	45.4	60.0	-14.6	Line
33	269.988k	45.4	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	51.3	66.0	-14.7	Line
34	28.369M	37.8	+0.2 +0.8	+0.2	+0.5	+5.7	+0.0	45.2	60.0	-14.8	Line
35	29.774M	37.7	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.2	60.0	-14.8	Line
36	15.178M	38.4	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	45.1	60.0	-14.9	Line
37	29.789M	37.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.9	60.0	-15.1	Line
38	29.406M	37.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.6	60.0	-15.4	Line
39	29.765M	37.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.6	60.0	-15.4	Line
40	26.339M	36.9	+0.2 +0.8	+0.2	+0.4	+5.7	+0.0	44.2	60.0	-15.8	Line
41	28.152M	36.8	+0.2 +0.8	+0.2	+0.5	+5.7	+0.0	44.2	60.0	-15.8	Line
42	28.297M	36.8	+0.2 +0.8	+0.2	+0.5	+5.7	+0.0	44.2	60.0	-15.8	Line
43	21.611M	36.9	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	44.1	60.0	-15.9	Line

44	29.668M	36.6	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.1	60.0	-15.9	Line
45	29.970M	36.6	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.1	60.0	-15.9	Line
46	549.963k	38.0	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	44.0	60.0	-16.0	Line
47	21.791M	36.8	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	44.0	60.0	-16.0	Line
48	29.752M	36.5	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.0	60.0	-16.0	Line
49	28.730M	36.3	+0.2 +0.8	+0.2	+0.5	+5.7	+0.0	43.7	60.0	-16.3	Line
50	29.587M	36.2	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	43.7	60.0	-16.3	Line

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**EN 55032 AC Mains Class A - Average**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Conducted Emissions**	Time:	3:56:19 PM
Tested by:	S. Yamamoto	Sequence#:	6
Software:	EMITest 5.03.02		230V 50Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 2			

Support Equipment:

Device	Manufacturer	Model #	S/N
Configuration 2			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 150kHz to 30MHz. RBW=VBW=9kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
Modification 1 was in place during testing.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 6 Date: 7/8/2016
EN 55032 AC Mains Class A - Average Test Lead: 230V 50Hz Neutral
dBµV
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Peak Readings
Average Readings
Software Version: 5.03.02
2 - EN 55032 AC Mains Class A - Quasi-peak
Readings
QP Readings
Ambient
1 - EN 55032 AC Mains Class A - Average

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T1	AN02343	High Pass Filter	HE9615-150K-50-720B	1/8/2015	1/8/2017
T2	ANP01910	Cable	RG-142	11/30/2015	11/30/2017
T3	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T4	ANP06085	Attenuator	SA18N10W-09	12/17/2014	12/17/2016
	AN00969A	50uH LISN-Line 1 (L1) (dB)	3816/2NM	3/12/2015	3/12/2017
T5	AN00969A	50uH LISN-Line 2 (L2) (dB)	3816/2NM	3/12/2015	3/12/2017

Measurement Data: Reading listed by margin. Test Lead: Neutral

#	Freq	Rdng	T1 T5	T2	T3	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV	dBµV	dB	Ant
1	17.695M	51.0	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.9	60.0	-2.1	Neutr
2	18.083M	50.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.7	60.0	-2.3	Neutr
3	18.002M	50.7	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.6	60.0	-2.4	Neutr
4	17.262M	50.5	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.4	60.0	-2.6	Neutr
5	157.272k	54.8	+1.3 +0.1	+0.0	+0.0	+5.7	+0.0	61.9	66.0	-4.1	Neutr
6	191.814k	55.1	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	61.0	66.0	-5.0	Neutr
7	168.180k	54.4	+0.3 +0.1	+0.0	+0.0	+5.7	+0.0	60.5	66.0	-5.5	Neutr
8	199.086k	54.1	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	60.0	66.0	-6.0	Neutr
9	18.949M	45.8	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	52.8	60.0	-7.2	Neutr
10	19.175M	45.5	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	52.5	60.0	-7.5	Neutr
11	208.176k	52.5	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	58.4	66.0	-7.6	Neutr
12	16.252M	45.5	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	52.4	60.0	-7.6	Neutr
13	204.540k	52.2	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	58.1	66.0	-7.9	Neutr
14	19.455M	44.2	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	51.2	60.0	-8.8	Neutr
15	19.527M	44.0	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	51.1	60.0	-8.9	Neutr
16	19.797M	43.6	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	50.7	60.0	-9.3	Neutr
17	19.689M	43.5	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	50.6	60.0	-9.4	Neutr

18	228.174k	50.5	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	56.4	66.0	-9.6	Neutr
19	220.902k	50.1	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	56.0	66.0	-10.0	Neutr
20	239.082k	50.0	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	55.9	66.0	-10.1	Neutr
21	20.041M	42.7	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	49.9	60.0	-10.1	Neutr
22	19.978M	41.8	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	49.0	60.0	-11.0	Neutr
23	20.077M	41.7	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	48.9	60.0	-11.1	Neutr
24	19.996M	41.6	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	48.8	60.0	-11.2	Neutr
25	20.104M	40.9	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	48.1	60.0	-11.9	Neutr
26	264.534k	48.0	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	53.9	66.0	-12.1	Neutr
27	15.449M	40.4	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	47.2	60.0	-12.8	Neutr
28	20.249M	39.7	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	46.9	60.0	-13.1	Neutr
29	13.328M	40.3	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.8	60.0	-13.2	Neutr
30	13.418M	40.3	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.8	60.0	-13.2	Neutr
31	13.689M	40.2	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	46.8	60.0	-13.2	Neutr
32	20.330M	39.5	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	46.7	60.0	-13.3	Neutr
33	27.430M	38.9	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	46.5	60.0	-13.5	Neutr
34	25.482M	38.7	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	46.1	60.0	-13.9	Neutr
35	29.764M	38.4	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	46.1	60.0	-13.9	Neutr
36	29.816M	38.3	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	46.0	60.0	-14.0	Neutr
37	29.895M	38.3	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	46.0	60.0	-14.0	Neutr
38	20.348M	38.5	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	45.7	60.0	-14.3	Neutr
39	27.927M	38.1	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	45.7	60.0	-14.3	Neutr
40	29.677M	38.0	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.7	60.0	-14.3	Neutr
41	29.777M	37.9	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.6	60.0	-14.4	Neutr
42	29.801M	37.9	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.6	60.0	-14.4	Neutr
43	13.761M	38.9	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	45.5	60.0	-14.5	Neutr

44	575.415k	39.5	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	45.5	60.0	-14.5	Neutr
45	29.856M	37.8	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.5	60.0	-14.5	Neutr
46	13.824M	38.9	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	45.5	60.0	-14.5	Neutr
47	29.836M	37.7	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.4	60.0	-14.6	Neutr
48	29.839M	37.7	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.4	60.0	-14.6	Neutr
49	15.286M	38.5	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	45.3	60.0	-14.7	Neutr
50	29.846M	37.6	+0.2 +1.1	+0.2	+0.5	+5.7	+0.0	45.3	60.0	-14.7	Neutr

Test Setup Photo(s)



Telecom Conducted Emissions

Test Notes: Conducted Disturbances at Mains Terminals, T-LISN method.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	NA
Test Method:	EN 55032	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares the EUT does not have telecommunication ports.		
Configuration:	NA		

Radiated Emissions

Test Notes: Radiated disturbances emanating from enclosure.

Test Data

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**EN 55032 Radiated Emissions Class A (3m)**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Maximized Emissions**	Time:	14:31:26
Tested by:	S. Yamamoto	Sequence#:	2
Software:	EMITest 5.03.02		

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 2			

Support Equipment:

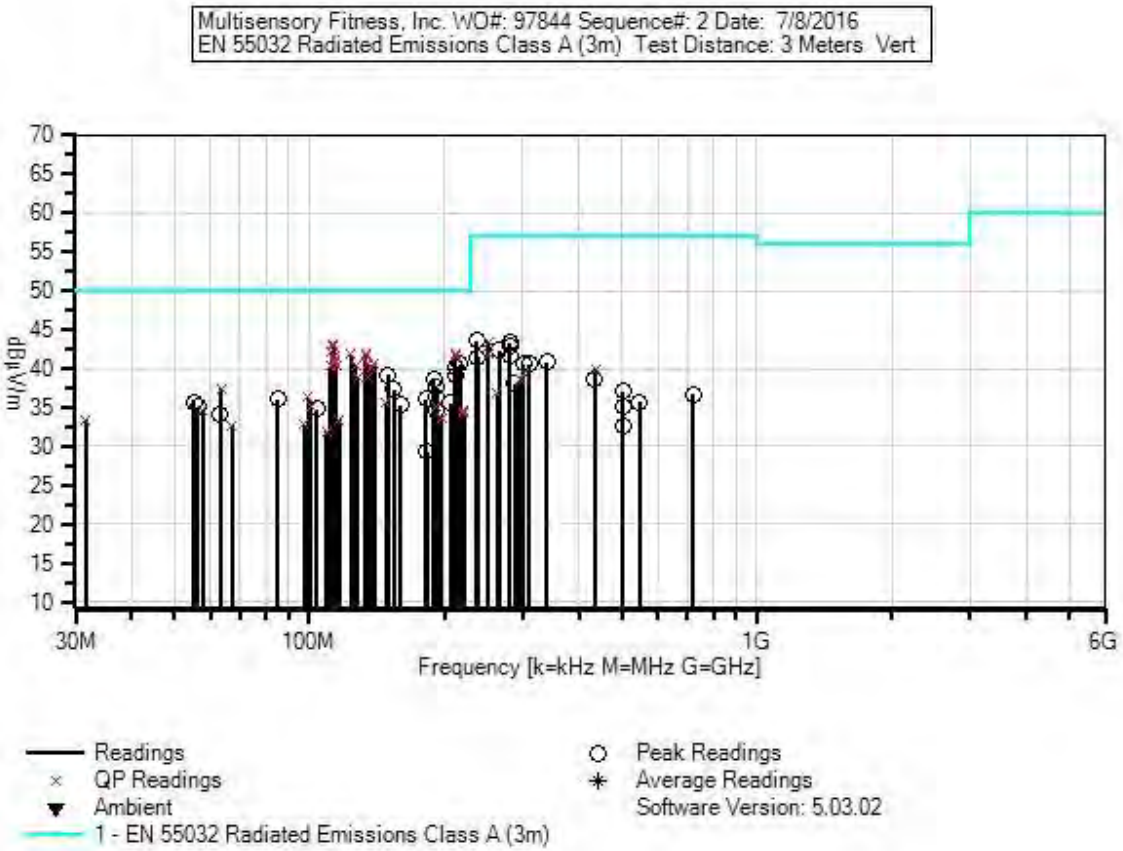
Device	Manufacturer	Model #	S/N
Configuration 2			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 30MHz to 1000MHz. RBW=VBW=120kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
The manufacturer declares that the highest frequency generated or used in the device is less than 108MHz.
Voltage to the EUT is 230Vac 50Hz.
Modification 1 was in place during testing.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 2 Date: 7/8/2016
EN 55032 Radiated Emissions Class A (3m) Test Distance: 3 Meters Vert
dBµV/m
70
65
60
55
50
45
40
35
30
25
20
15
10
30M
100M
1G
6G
Frequency [k=kHz M=MHz G=GHz]
Readings
QP Readings
Ambient
1 - EN 55032 Radiated Emissions Class A (3m)
Peak Readings
Average Readings
Software Version: 5.03.02

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
T1	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T2	ANP04382	Cable	LDF-50	6/6/2016	6/6/2018
T3	ANP05555	Cable	RG223/U	4/5/2016	4/5/2018
T4	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T5	AN00010	Preamp	8447D	3/14/2016	3/14/2018
T6	AN01992	Biconilog Antenna	CBL6111C	12/4/2014	12/4/2016
T7	ANP05283	Attenuator	ATT-0218-06-NNN-02	5/5/2016	5/5/2018

Measurement Data: Reading listed by margin. Test Distance: 3 Meters

#	Freq	Rdng	T1 T5	T2 T6	T3 T7	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV/m	dBµV/m	dB	Ant
1	113.235M	51.1	+0.0	+1.1	+0.1	+1.1	+0.0	43.3	50.0	-6.7	Vert
	QP		-27.0	+11.1	+5.8						
^	113.235M	53.3	+0.0	+1.1	+0.1	+1.1	+0.0	45.5	50.0	-4.5	Vert
			-27.0	+11.1	+5.8						
3	112.970M	50.6	+0.0	+1.1	+0.1	+1.1	+0.0	42.8	50.0	-7.2	Vert
	QP		-27.0	+11.1	+5.8						
^	112.970M	51.3	+0.0	+1.1	+0.1	+1.1	+0.0	43.5	50.0	-6.5	Vert
			-27.0	+11.1	+5.8						
5	112.170M	49.9	+0.0	+1.1	+0.1	+1.1	+0.0	42.0	50.0	-8.0	Vert
	QP		-27.0	+11.0	+5.8						
^	112.170M	51.7	+0.0	+1.1	+0.1	+1.1	+0.0	43.8	50.0	-6.2	Vert
			-27.0	+11.0	+5.8						
7	133.260M	49.3	+0.0	+1.2	+0.1	+1.2	+0.0	42.0	50.0	-8.0	Horiz
	QP		-27.0	+11.4	+5.8						
^	133.260M	53.0	+0.0	+1.2	+0.1	+1.2	+0.0	45.7	50.0	-4.3	Horiz
			-27.0	+11.4	+5.8						
9	211.350M	49.7	+0.0	+1.5	+0.2	+1.5	+0.0	42.0	50.0	-8.0	Horiz
	QP		-26.7	+10.0	+5.8						
^	211.350M	51.4	+0.0	+1.5	+0.2	+1.5	+0.0	43.7	50.0	-6.3	Horiz
			-26.7	+10.0	+5.8						
11	123.600M	49.2	+0.0	+1.1	+0.1	+1.1	+0.0	41.8	50.0	-8.2	Vert
	QP		-27.0	+11.5	+5.8						
^	123.600M	55.1	+0.0	+1.1	+0.1	+1.1	+0.0	47.7	50.0	-2.3	Vert
			-27.0	+11.5	+5.8						
13	133.677M	48.9	+0.0	+1.2	+0.1	+1.2	+0.0	41.6	50.0	-8.4	Vert
	QP		-27.0	+11.4	+5.8						
^	133.677M	54.5	+0.0	+1.2	+0.1	+1.2	+0.0	47.2	50.0	-2.8	Vert
			-27.0	+11.4	+5.8						
15	211.552M	49.3	+0.0	+1.5	+0.2	+1.5	+0.0	41.6	50.0	-8.4	Horiz
	QP		-26.7	+10.0	+5.8						
^	211.552M	53.8	+0.0	+1.5	+0.2	+1.5	+0.0	46.1	50.0	-3.9	Horiz
			-26.7	+10.0	+5.8						

17	113.540M QP	49.3	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	41.5	50.0	-8.5	Vert
18	113.450M QP	48.9	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	41.1	50.0	-8.9	Vert
^	113.540M	53.2	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	45.4	50.0	-4.6	Vert
20	113.405M QP	48.6	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	40.8	50.0	-9.2	Vert
^	113.405M	54.9	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	47.1	50.0	-2.9	Vert
^	113.450M	54.1	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	46.3	50.0	-3.7	Vert
23	217.475M	47.8	+0.0 -26.7	+1.5 +10.5	+0.2 +5.8	+1.5	+0.0	40.6	50.0	-9.4	Vert
24	139.560M QP	47.7	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.5	50.0	-9.5	Vert
^	139.560M	49.2	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	42.0	50.0	-8.0	Vert
26	135.360M QP	47.8	+0.0 -27.0	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.5	50.0	-9.5	Horiz
27	127.040M QP	47.8	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	40.4	50.0	-9.6	Horiz
^	127.040M	52.7	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	45.3	50.0	-4.7	Horiz
29	111.645M QP	48.3	+0.0 -27.1	+1.1 +11.0	+0.1 +5.8	+1.1	+0.0	40.3	50.0	-9.7	Vert
^	111.645M	53.1	+0.0 -27.1	+1.1 +11.0	+0.1 +5.8	+1.1	+0.0	45.1	50.0	-4.9	Vert
31	115.020M QP	48.0	+0.0 -27.0	+1.1 +11.2	+0.1 +5.8	+1.1	+0.0	40.3	50.0	-9.7	Vert
^	115.020M	51.8	+0.0 -27.0	+1.1 +11.2	+0.1 +5.8	+1.1	+0.0	44.1	50.0	-5.9	Vert
33	138.085M QP	47.4	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.2	50.0	-9.8	Vert
^	138.085M	52.4	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	45.2	50.0	-4.8	Vert
35	213.008M	47.7	+0.0 -26.7	+1.5 +10.1	+0.2 +5.8	+1.5	+0.0	40.1	50.0	-9.9	Horiz
36	149.675M	46.3	+0.0 -26.9	+1.3 +11.2	+0.2 +5.8	+1.3	+0.0	39.2	50.0	-10.8	Vert
37	212.837M	46.8	+0.0 -26.7	+1.5 +10.1	+0.2 +5.8	+1.5	+0.0	39.2	50.0	-10.8	Horiz
38	128.300M QP	46.3	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	38.9	50.0	-11.1	Horiz
^	128.300M	51.8	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	44.4	50.0	-5.6	Horiz
^	128.300M	51.5	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	44.1	50.0	-5.9	Horiz

41	135.400M	46.1	+0.0	+1.2	+0.1	+1.2	+0.0	38.8	50.0	-11.2	Horiz
	QP		-27.0	+11.4	+5.8						
^	135.360M	54.5	+0.0	+1.2	+0.1	+1.2	+0.0	47.2	50.0	-2.8	Horiz
			-27.0	+11.4	+5.8						
^	135.400M	50.8	+0.0	+1.2	+0.1	+1.2	+0.0	43.5	50.0	-6.5	Horiz
			-27.0	+11.4	+5.8						
44	190.075M	47.5	+0.0	+1.5	+0.2	+1.4	+0.0	38.7	50.0	-11.3	Vert
			-26.8	+9.1	+5.8						
45	192.075M	46.4	+0.0	+1.5	+0.2	+1.4	+0.0	37.6	50.0	-12.4	Vert
			-26.8	+9.1	+5.8						
46	63.550M	51.0	+0.0	+0.8	+0.1	+0.7	+0.0	37.4	50.0	-12.6	Vert
	QP		-27.2	+6.2	+5.8						
^	63.550M	54.4	+0.0	+0.8	+0.1	+0.7	+0.0	40.8	50.0	-9.2	Vert
			-27.2	+6.2	+5.8						
48	154.485M	44.6	+0.0	+1.3	+0.2	+1.3	+0.0	37.3	50.0	-12.7	Horiz
			-26.9	+11.0	+5.8						
49	235.939M	49.3	+0.0	+1.6	+0.2	+1.6	+0.0	43.6	57.0	-13.4	Horiz
			-26.7	+11.8	+5.8						
50	250.686M	48.0	+0.0	+1.6	+0.2	+1.7	+0.0	43.4	57.0	-13.6	Vert
	QP		-26.6	+12.7	+5.8						
^	250.686M	49.8	+0.0	+1.6	+0.2	+1.7	+0.0	45.2	57.0	-11.8	Vert
			-26.6	+12.7	+5.8						
^	250.672M	47.4	+0.0	+1.6	+0.2	+1.7	+0.0	42.8	57.0	-14.2	Vert
			-26.6	+12.7	+5.8						
53	99.278M	45.4	+0.0	+1.0	+0.1	+1.0	+0.0	36.3	50.0	-13.7	Vert
	QP		-27.1	+10.1	+5.8						
^	99.278M	48.3	+0.0	+1.0	+0.1	+1.0	+0.0	39.2	50.0	-10.8	Vert
			-27.1	+10.1	+5.8						
55	191.675M	45.1	+0.0	+1.5	+0.2	+1.4	+0.0	36.3	50.0	-13.7	Vert
			-26.8	+9.1	+5.8						
56	280.190M	47.3	+0.0	+1.7	+0.2	+1.8	+0.0	43.3	57.0	-13.7	Horiz
			-26.6	+13.1	+5.8						
57	85.050M	47.0	+0.0	+1.0	+0.1	+0.9	+0.0	36.1	50.0	-13.9	Vert
			-27.1	+8.4	+5.8						
58	182.350M	44.9	+0.0	+1.4	+0.2	+1.4	+0.0	36.0	50.0	-14.0	Vert
			-26.8	+9.1	+5.8						
59	280.169M	46.9	+0.0	+1.7	+0.2	+1.8	+0.0	42.9	57.0	-14.1	Horiz
			-26.6	+13.1	+5.8						
60	148.350M	43.1	+0.0	+1.2	+0.1	+1.2	+0.0	35.7	50.0	-14.3	Vert
	QP		-26.9	+11.2	+5.8						
^	148.350M	48.6	+0.0	+1.2	+0.1	+1.2	+0.0	41.2	50.0	-8.8	Vert
			-26.9	+11.2	+5.8						
62	55.300M	48.1	+0.0	+0.8	+0.1	+0.7	+0.0	35.6	50.0	-14.4	Vert
			-27.2	+7.3	+5.8						
63	206.453M	43.7	+0.0	+1.5	+0.2	+1.4	+0.0	35.5	50.0	-14.5	Horiz
			-26.7	+9.6	+5.8						
64	265.445M	46.7	+0.0	+1.7	+0.2	+1.7	+0.0	42.4	57.0	-14.6	Horiz
			-26.6	+12.9	+5.8						
65	159.610M	42.8	+0.0	+1.3	+0.2	+1.3	+0.0	35.3	50.0	-14.7	Horiz
			-26.9	+10.8	+5.8						
66	56.287M	47.8	+0.0	+0.8	+0.1	+0.7	+0.0	35.1	50.0	-14.9	Vert
			-27.2	+7.1	+5.8						

67	100.570M	44.0	+0.0	+1.0	+0.1	+1.0	+0.0	35.0	50.0	-15.0	Vert
	QP		-27.1	+10.2	+5.8						
^	100.570M	49.5	+0.0	+1.0	+0.1	+1.0	+0.0	40.5	50.0	-9.5	Vert
			-27.1	+10.2	+5.8						
69	192.370M	43.8	+0.0	+1.5	+0.2	+1.4	+0.0	35.0	50.0	-15.0	Vert
	QP		-26.8	+9.1	+5.8						
^	192.370M	49.8	+0.0	+1.5	+0.2	+1.4	+0.0	41.0	50.0	-9.0	Vert
			-26.8	+9.1	+5.8						
71	104.009M	43.5	+0.0	+1.0	+0.1	+1.0	+0.0	34.8	50.0	-15.2	Vert
			-27.1	+10.5	+5.8						
72	250.702M	46.4	+0.0	+1.6	+0.2	+1.7	+0.0	41.8	57.0	-15.2	Horiz
	QP		-26.6	+12.7	+5.8						
^	250.702M	49.0	+0.0	+1.6	+0.2	+1.7	+0.0	44.4	57.0	-12.6	Horiz
			-26.6	+12.7	+5.8						
74	57.613M	47.7	+0.0	+0.8	+0.1	+0.7	+0.0	34.7	50.0	-15.3	Horiz
	QP		-27.2	+6.8	+5.8						
^	57.613M	48.8	+0.0	+0.8	+0.1	+0.7	+0.0	35.8	50.0	-14.2	Horiz
			-27.2	+6.8	+5.8						
76	280.199M	45.7	+0.0	+1.7	+0.2	+1.8	+0.0	41.7	57.0	-15.3	Horiz
			-26.6	+13.1	+5.8						
77	191.704M	43.4	+0.0	+1.5	+0.2	+1.4	+0.0	34.6	50.0	-15.4	Horiz
			-26.8	+9.1	+5.8						
78	220.050M	41.7	+0.0	+1.5	+0.2	+1.5	+0.0	34.6	50.0	-15.4	Horiz
	QP		-26.7	+10.6	+5.8						
^	220.050M	46.2	+0.0	+1.5	+0.2	+1.5	+0.0	39.1	50.0	-10.9	Horiz
			-26.7	+10.6	+5.8						
80	238.100M	47.0	+0.0	+1.6	+0.2	+1.6	+0.0	41.5	57.0	-15.5	Horiz
			-26.6	+11.9	+5.8						
81	221.203M	41.4	+0.0	+1.5	+0.2	+1.5	+0.0	34.4	50.0	-15.6	Horiz
	QP		-26.7	+10.7	+5.8						
^	221.203M	43.5	+0.0	+1.5	+0.2	+1.5	+0.0	36.5	50.0	-13.5	Horiz
			-26.7	+10.7	+5.8						
83	63.088M	47.6	+0.0	+0.8	+0.1	+0.7	+0.0	34.0	50.0	-16.0	Horiz
			-27.2	+6.2	+5.8						
84	339.157M	42.9	+0.0	+2.0	+0.3	+2.0	+0.0	40.8	57.0	-16.2	Horiz
			-26.8	+14.6	+5.8						
85	197.320M	42.4	+0.0	+1.5	+0.2	+1.4	+0.0	33.7	50.0	-16.3	Vert
	QP		-26.7	+9.1	+5.8						
^	197.320M	47.6	+0.0	+1.5	+0.2	+1.4	+0.0	38.9	50.0	-11.1	Vert
			-26.7	+9.1	+5.8						
87	309.669M	44.1	+0.0	+1.8	+0.2	+1.8	+0.0	40.6	57.0	-16.4	Horiz
			-26.7	+13.6	+5.8						
88	300.710M	44.2	+0.0	+1.8	+0.2	+1.8	+0.0	40.5	57.0	-16.5	Horiz
			-26.6	+13.3	+5.8						
89	31.650M	36.4	+0.0	+0.5	+0.0	+0.5	+0.0	33.5	50.0	-16.5	Vert
	QP		-27.2	+17.5	+5.8						
^	31.650M	40.3	+0.0	+0.5	+0.0	+0.5	+0.0	37.4	50.0	-12.6	Vert
			-27.2	+17.5	+5.8						
91	116.338M	41.0	+0.0	+1.1	+0.1	+1.1	+0.0	33.4	50.0	-16.6	Horiz
	QP		-27.0	+11.3	+5.8						
^	116.338M	47.3	+0.0	+1.1	+0.1	+1.1	+0.0	39.7	50.0	-10.3	Horiz
			-27.0	+11.3	+5.8						

93	436.000M	40.0	+0.0	+2.2	+0.3	+2.3	+0.0	40.0	57.0	-17.0	Horiz
	QP		-27.6	+17.0	+5.8						
	^ 436.000M	44.6	+0.0	+2.2	+0.3	+2.3	+0.0	44.6	57.0	-12.4	Horiz
			-27.6	+17.0	+5.8						
95	97.296M	42.1	+0.0	+1.0	+0.1	+1.0	+0.0	32.8	50.0	-17.2	Vert
	QP		-27.1	+9.9	+5.8						
	^ 97.296M	47.1	+0.0	+1.0	+0.1	+1.0	+0.0	37.8	50.0	-12.2	Vert
			-27.1	+9.9	+5.8						
97	67.250M	46.3	+0.0	+0.8	+0.1	+0.8	+0.0	32.7	50.0	-17.3	Vert
	QP		-27.2	+6.1	+5.8						
	^ 67.250M	50.7	+0.0	+0.8	+0.1	+0.8	+0.0	37.1	50.0	-12.9	Vert
			-27.2	+6.1	+5.8						
99	301.360M	43.1	+0.0	+1.8	+0.2	+1.8	+0.0	39.4	57.0	-17.6	Horiz
	QP		-26.6	+13.3	+5.8						
	^ 301.360M	48.9	+0.0	+1.8	+0.2	+1.8	+0.0	45.2	57.0	-11.8	Horiz
			-26.6	+13.3	+5.8						
101	109.000M	40.3	+0.0	+1.0	+0.1	+1.0	+0.0	31.9	50.0	-18.1	Vert
	QP		-27.1	+10.8	+5.8						
	^ 109.000M	47.2	+0.0	+1.0	+0.1	+1.0	+0.0	38.8	50.0	-11.2	Vert
			-27.1	+10.8	+5.8						
103	434.000M	38.8	+0.0	+2.2	+0.3	+2.3	+0.0	38.7	57.0	-18.3	Vert
			-27.6	+16.9	+5.8						
104	294.919M	42.4	+0.0	+1.8	+0.2	+1.8	+0.0	38.6	57.0	-18.4	Horiz
	QP		-26.6	+13.2	+5.8						
	^ 294.928M	47.2	+0.0	+1.8	+0.2	+1.8	+0.0	43.4	57.0	-13.6	Horiz
			-26.6	+13.2	+5.8						
	^ 294.919M	46.0	+0.0	+1.8	+0.2	+1.8	+0.0	42.2	57.0	-14.8	Horiz
			-26.6	+13.2	+5.8						
107	288.000M	41.9	+0.0	+1.8	+0.2	+1.8	+0.0	38.1	57.0	-18.9	Vert
			-26.6	+13.2	+5.8						
108	501.379M	35.9	+0.0	+2.4	+0.4	+2.5	+0.0	37.0	57.0	-20.0	Horiz
			-28.0	+18.0	+5.8						
109	262.177M	41.1	+0.0	+1.7	+0.2	+1.7	+0.0	36.8	57.0	-20.2	Horiz
	QP		-26.6	+12.9	+5.8						
	^ 262.177M	47.7	+0.0	+1.7	+0.2	+1.7	+0.0	43.4	57.0	-13.6	Horiz
			-26.6	+12.9	+5.8						
111	720.101M	31.8	+0.0	+2.9	+0.4	+3.1	+0.0	36.7	57.0	-20.3	Horiz
			-28.0	+20.7	+5.8						
112	183.360M	38.1	+0.0	+1.4	+0.2	+1.4	+0.0	29.2	50.0	-20.8	Horiz
			-26.8	+9.1	+5.8						
113	545.592M	33.6	+0.0	+2.5	+0.3	+2.6	+0.0	35.6	57.0	-21.4	Horiz
			-28.1	+18.9	+5.8						
114	501.372M	34.1	+0.0	+2.4	+0.4	+2.5	+0.0	35.2	57.0	-21.8	Vert
			-28.0	+18.0	+5.8						
115	501.375M	31.4	+0.0	+2.4	+0.4	+2.5	+0.0	32.5	57.0	-24.5	Horiz
			-28.0	+18.0	+5.8						

Test Setup Photo(s)





EN 61000-3-2 HARMONIC EMISSIONS

Test Notes: Conducted disturbances of current harmonics emanating from equipment power supply to low-voltage public power networks.

Test Setup/Conditions			
Test Location:	Brea	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-3-2	Test Date(s):	7/11/2016
Test Level:	Harmonic Emissions		
Declarations:	Modification 1 was in place during testing.		
Configuration:	3		

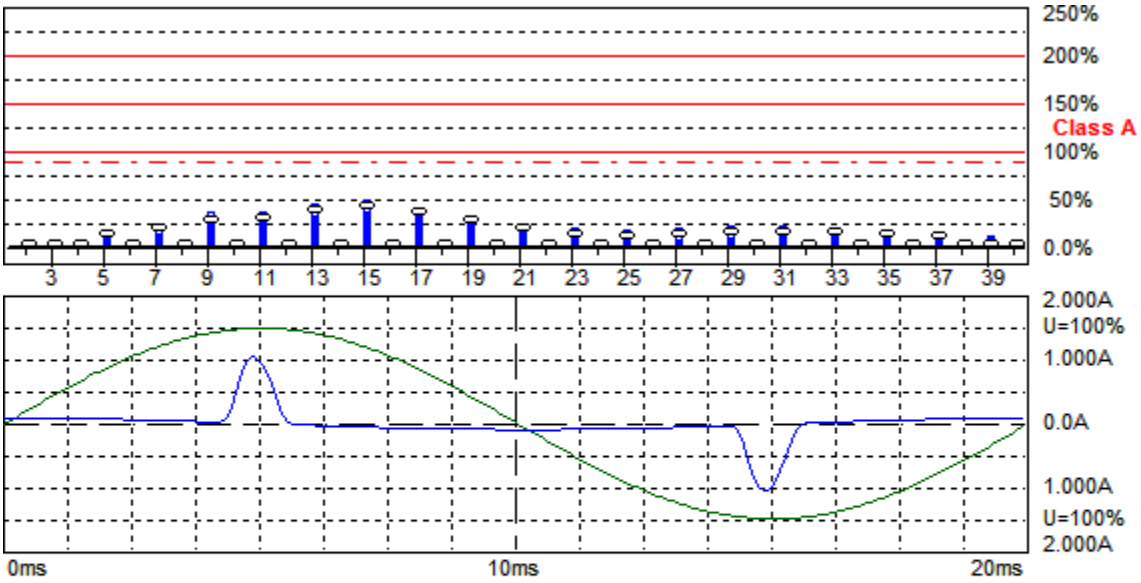
Test Equipment					
Asset #	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
02476	AC Power Source	California Instruments	1251WP	5/20/2015	5/20/2017
02890	Flicker & Harmonics Meter	EMC Partner AG	HAR1000-1P	5/29/2015	5/29/2017

Harmonics Emissions Results Table		
Standard	**Test**	**Pass/Fail**
EN 61000-3-2	Harmonic Emissions	Pass

Harmonic Emission - EN 61000-3-2

Multisensory Fitness, Inc.

Operator	S. Yamamoto
Unit	Configuration 3
Serial Number	None
Remarks	Game 600. Modification 1 was in place during testing.



Harmonic Emission - IEC 61000-3-2 , EN 61000-3-2 , (EN60555-2) 7/11/2016 8:31:01 PM

Urms =	230.1	V	P =	25.52	W	THC =	0.232	A	
Irms =	0.260	A	pf =	0.427					

Range: 2 A
V-nom: 230 V
TestTime: 5 min (100%)

Configuration 1

Test completed, Result: PASSED

Game 600

HAR-1000 EMC-Partner

Full Bar : Actual Values
Empty Bar : Maximum Values
Blue : Current , Green : Voltage , Red : Failed

Measurement

```
Multisensory Fitness, Inc.                    Date: 7/11/2016 8:31:01 PM V4.21

Operator                        S. Yamamoto
Unit                            Configuration 3
Serial Number                   None
Remarks                         Game 600.  Modification 1 was in place
during testing.

Urms =  230.1V  Freq =  50.065  Range:  2 A
Irms =  0.260A  Ipk  =  1.104A  cf    =  4.252
P    =  25.52W  S    =  59.77VA pf    =  0.427
THDi =  180 %   THDu =  0.10 %  Class A

Test - Time :   5min    ( 100 %)

Test completed, Result: PASSED
```

Test Setup Photo



EN 61000-3-3 VOLTAGE FLUCTUATION AND FLICKER EMISSIONS

Test Notes: Conducted disturbances of voltage fluctuations and flicker emanating from equipment power supply to low-voltage public power networks.

Test Setup/Conditions			
Test Location:	Brea	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-3-3	Test Date(s):	7/11/2016
Test Level:	Flicker Emissions		
Declarations:	Modification 1 was in place during testing.		
Configuration:	3		

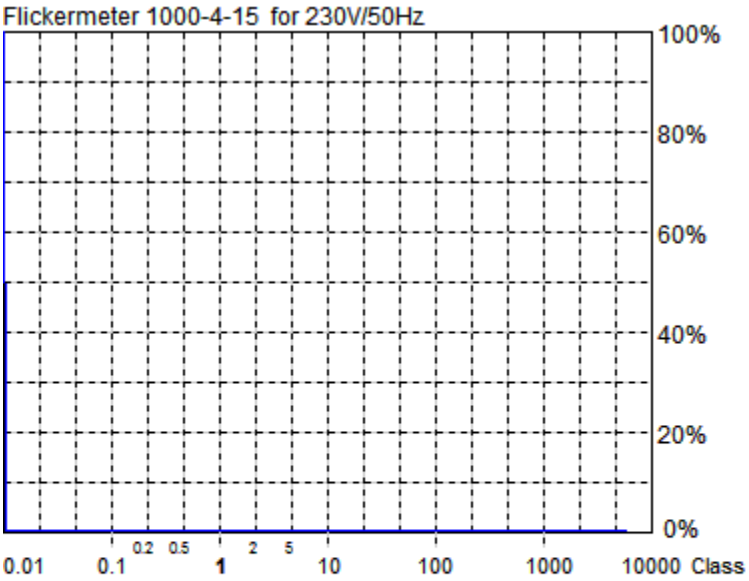
Test Equipment					
Asset #	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
02476	AC Power Source	California Instruments	1251WP	5/20/2015	5/20/2017
02890	Flicker & Harmonics Meter	EMC Partner AG	HAR1000-1P	5/29/2015	5/29/2017

Voltage Fluctuation and Flicker Emissions Results Table		
Standard	**Test**	**Pass/Fail**
EN 61000-3-3	Voltage Fluctuations and Flicker Emissions	Pass

Flicker Emission - EN 61000-3-3

Multisensory Fitness, Inc.

Operator	S. Yamamoto
Unit	Configuration 3
Serial Number	None
Remarks	Game 600. Modification 1 was in place during testing.



Actual Flicker (Fli):	**0.00**
Short-term Flicker (Pst):	**0.07**
Limit (Pst):	1.00
Long-term Flicker (Plt):	**0.07**
Limit (Plt):	0.65
Maximum Relative Volt. Change (dmax):	**0.00%**
Limit (dmax):	4.00%
Relative Steady-state Voltage Change (dc):	**0.11%**
Limit (dc):	3.00%
Maximum Interval exceeding 3.00% (dt):	**0.00ms**
Limit (dt>Lim):	200ms

Flicker Emission - IEC 61000-3-3 , EN 61000-3-3

7/11/2016 8:43:13 PM

Urms =	229.9	V	P =	32.98	W
Irms =	0.313	A	pf =	0.458	

Range:	2 A
V-nom:	230 V
TestTime:	10 min (100%)

Configuration 1

Test completed, Result: PASSED

Game 600

HAR-1000 EMC-Partner

Full Bar : Actual Values
Empty Bar : Maximum Values
Circles : Average Values
Blue : Current , Green : Voltage , Red : Failed

Measurement

```
Multisensory Fitness, Inc.                    Date: 7/11/2016 8:43:13 PM V4.21

Operator                        S. Yamamoto
Unit                            Configuration 3
Serial Number                   none
Remarks                         Game 600. Modification 1 was in place during
testing.

Urms =  229.9V  Freq =  50.065  Range:  2 A
Irms =  0.313A  Ipk  =  1.280A  cf   =  4.084
P    =  32.98W  S    =  72.07VA pf   =  0.458

Test - Time :   1 x 10min = 10min      ( 100 %)

LIN (Line Impedance Network) :  L: 0.24ohm +j0.15ohm N: 0.16ohm +j0.10ohm

Limits :        Plt   :  0.65    Pst    : 1.00
                dmax  :  4.00 %  dc     : 3.00 %
                dtLim:   3.00 %  dt>Lim: 200ms

Test completed, Result: PASSED
```

Test Setup Photo



EN 55024 IMMUNITY

Immunity Performance Criteria Definitions

Designation	Definition
Standard / Mfg.	**Standard Requirement / Manufacturer's Declaration**
A	During and after the test the EUT shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a minimum performance level specified by the manufacturer when the EUT is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the EUT if used as intended.
B	After the test, the EUT shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the EUT is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test. If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the EUT if used as intended.
C	During and after testing, a temporary loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls or cycling of the power to the EUT by the user in accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.
MA	During the test, unintended continuous flashing on/off of LEDs, unexpected increment in score, loss of functionality of the sensors, is considered normal functionality, hence is not considered a failure unless the device lost totally functionality when the interference is removed. The equipment is considered to fail only if after the immunity threat is removed the system cannot be restarted with a power cycle.
MB	None declared.
MC	None declared.

Immunity Setup and Monitoring

Equipment Setup: The equipment under test (EUT) is setup to run either game 600 or game 102. Before, during, and after the test at least one sensor is pushed to check for functionality of the EUT sensors. The display is being monitored for score or timing as well as the sound from the speaker.

Degradation Criteria: If a sensor push is not registered, if the score automatically changes, or the timing stops or increments at an abnormally rapid rate.

Monitoring: The EUT displays, LEDs, and speaker are being monitored either locally or using a video camera and remote monitor.

Electrostatic Discharge

Test Notes: Immunity threat to simulate static electrical charges directly from an operator or from adjacent objects.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-2	Test Date(s):	7/11/2016
Test Level:	Contact: +/- 2 kV, 4 kV Air: +/- 2 kV, 4 kV, 8 kV		
Declarations:	Modification 1 was in place during testing. The manufacturer declares the EUT is floor standing.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	20	Relative Humidity (%):	40	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
01650	ESD Simulator	Keytek	MZ-15/ec	11/9/2015	11/9/2016

ESD Contact Discharge Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
VCP	Degradation requiring user intervention	All LED's turned off and the sensors did not respond to a push. It was necessary to recycle power to the unit in order for the equipment to function normally again.
Front	Degradation requiring user intervention	All LED's turned off and the sensors did not respond to a push. It was necessary to recycle power to the unit in order for the equipment to function normally again.
Back	NA	Not accessible after installation
Left Side	Degradation requiring user intervention	All LED's turned off and the sensors did not respond to a push. It was necessary to recycle power to the unit in order for the equipment to function normally again.
Right Side	Degradation requiring user intervention	All LED's turned off and the sensors did not respond to a push. It was necessary to recycle power to the unit in order for the equipment to function normally again.
Top	Degradation requiring user intervention	All LED's turned off and the sensors did not respond to a push. It was necessary to recycle power to the unit in order for the equipment to function normally again.

ESD Air Discharge Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
Front	No discharge	
Back	NA	Not accessible after installation
Left Side	No discharge	
Right Side	No discharge	
Top	No discharge	

ESD Test Summary		
Standard Minimum Performance Criteria:	B	
For this test, the EUT met performance criteria:	MA	Result: Pass

Test Setup Photo(s)

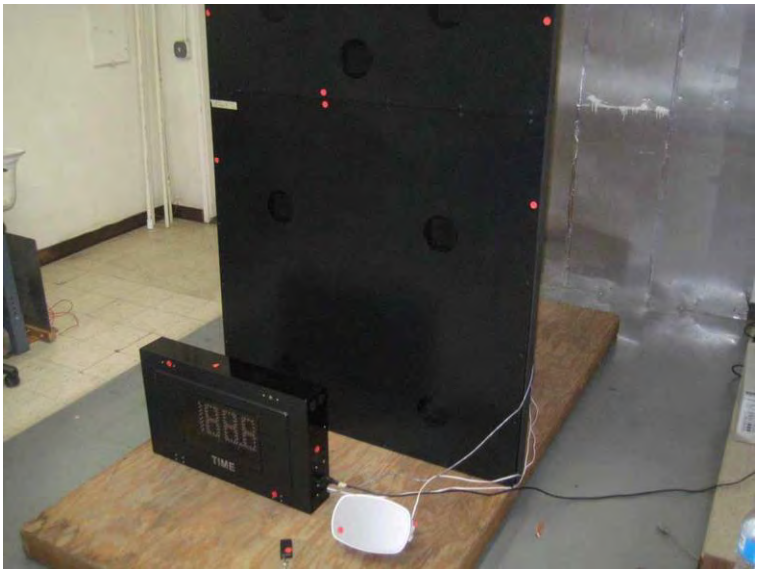
Red = Contact Discharge
Green = Air Discharge






SCORE
TIME


TIME


TIME

Radiated Immunity

Test Notes: Immunity threat to simulate radiated electro-magnetic fields.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-3	Test Date(s):	7/11/2016
Test Level:	3V/m 1kHz sine 80% AM		
Declarations:	Modification 1 was in place during testing.		
Configuration:	A		

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
BC 61000 4 3	Chamber Cal	CKC	Brea	1/6/2015	1/6/2017
03354	Log Periodic Antenna	Amplifier Research	ATR80M6G	1/6/2015	1/6/2017
00627	Amplifier	AR	50W1000A	1/7/2015	1/7/2017
01612	Directional Coupler	Werlatone	C2630	7/6/2015	7/6/2017
02467	Spectrum Analyzer	HP	E7405A	5/10/2016	5/10/2017
00636	Field Monitor	AR	FM2000	10/22/2015	10/22/2017
00638	Field Probe	AR	FP2000	10/22/2015	10/22/2017
P04270	Cable	Andrew	Hardline	NCR	NCR
P05008	Cable	Andrew	Hardline	1/6/2015	1/6/2017
P06139	Cable	Huber Suhner	Sucoflex 104A	1/7/2015	1/7/2017
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017

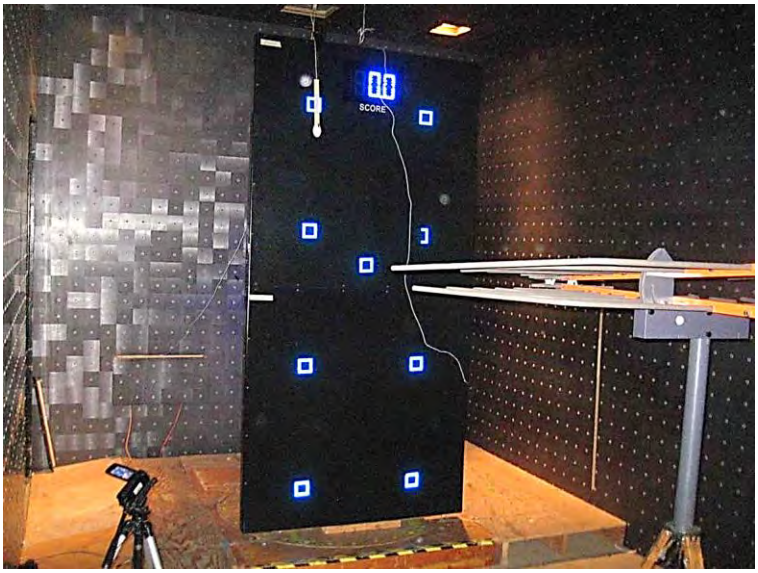
NCR = No Calibration Required

Radiated Immunity Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
Front	Degradation requiring user intervention	Unintended continuous flashing on/off of LEDs, unexpected increment in score, loss of functionality of the sensors. After the immunity threat is removed the system can be restarted and function normally with a power cycle.
Back	Degradation requiring user intervention	Unintended continuous flashing on/off of LEDs, unexpected increment in score, loss of functionality of the sensors. After the immunity threat is removed the system can be restarted and function normally with a power cycle.
Left Side	Degradation requiring user intervention	Unintended continuous flashing on/off of LEDs, unexpected increment in score, loss of functionality of the sensors. After the immunity threat is removed the system can be restarted and function normally with a power cycle.
Right Side	Degradation requiring user intervention	Unintended continuous flashing on/off of LEDs, unexpected increment in score, loss of functionality of the sensors. After the immunity threat is removed the system can be restarted and function normally with a power cycle.

Frequency Range	Test Distance (m)
80MHz to 1000MHz	1.5

Radiated Immunity Test Summary		
Standard Minimum Performance Criteria:	A	
For this test, the EUT met performance criteria:	MA	Result: Pass

Test Setup Photo(s)





Electrical Fast Transients / Bursts – AC/DC Mains Power Lines

Test Notes: Immunity threat to simulate electrical fast transient signals and bursts.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-4	Test Date(s):	7/11/2016
Test Level:	AC Mains: 1kV DC Mains: NA		
Declarations:	Modification 1 was in place during testing. DC Mains: The manufacturer declares that the EUT does not operate from DC mains.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	20	Relative Humidity (%):	40	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00844	Surge/EFT Generator	EM Test	USC 500M4	8/18/2015	8/18/2016

EFT/B AC Mains Power Lines Results Table		
Mode of operation:	Normal Operation	
Coupling Mode	**Test Results**	**Test Observations**
Line to Ground	No degradation	
Neutral to Ground	No degradation	
Protective Earth to Ground	No degradation	
Line/Neutral/Protective Earth to Ground.	No degradation	

EFT/B Test Summary		
Standard Minimum Performance Criteria:	B	
For this test, the EUT met performance criteria:	A	Result: Pass

Test Setup Photo(s)



Electrical Fast Transients / Bursts - IO Lines

Test Notes: Immunity threat to simulate electrical fast transient signals and bursts.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-4	Test Date(s):	7/11/2016
Test Level:	0.5kV		
Declarations:	Modification 1 was in place during testing.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	20	Relative Humidity (%):	40	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset #	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00844	Surge/EFT Generator	EM Test	USC 500M4	8/18/2015	8/18/2016
03422	Transient Event Simulator	EMTest	HFK	7/30/2015	7/30/2016

EFT/B IO Lines Results Table			
Mode of operation:	Normal Operation		
Port Tested		**Test Results**	**Test Observations**
Panel cable		No degradation	
Audio cable		No degradation	

Test Summary		
Standard Minimum Performance Criteria:	B	
For this test, the EUT met performance criteria:	A	Result: Pass

Test Setup Photo(s)



Surge AC / DC Mains

Test Notes: Immunity threat to simulate unidirectional surges caused by over voltages from switching and lightning transients.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-5	Test Date(s):	7/11/2016
Test Level:	AC Mains: 1/2 kV DC Mains: NA		
Declarations:	Modification 1 was in place during testing. DC Mains: The manufacturer declares that the EUT does not operate from DC mains.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	20	Relative Humidity (%):	40	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00844	Surge/EFT Generator	EM Test	USC 500M4	8/18/2015	8/18/2016

Surge AC Mains Results		
Mode of operation:	Normal Operation	
Coupling Mode	**Test Results**	**Test Observations**
Line-to-Line	No degradation	
Line-to-Ground	No degradation	
Neutral-to-Ground	No degradation	

Surge Test Summary		
Standard Minimum Performance Criteria:	B	
For this test, the EUT met performance criteria:	A	Result: Pass

Test Setup Photo(s)



Surge IO Lines

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	NA
Test Method:	EN 61000-4-5	Test Date(s):	NA
Test Level:	NA		
Declarations:	IO Lines: The manufacturer declares that the equipment has no ports which connect directly to outdoor cables.		
Configuration:	NA		

Conducted Immunity

Test Notes: Immunity threat to simulate low frequency radiated electro-magnetic fields having at least one conducting cable.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-6	Test Date(s):	7/11/2016
Test Level:	AC Mains: 3Vrms IO Lines: 3Vrms DC Mains: NA		
Declarations:	Modification 1 was in place during testing. DC Mains: The manufacturer declares that the EUT does not operate from DC mains.		
Configuration:	A		

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
03049	Coupling Decoupling Network	FCC	FCC-801-M3-25	1/5/2015	1/5/2017
00628	Amplifier	AR	100A100	1/7/2015	1/7/2017
01612	Directional Coupler	Werlatone	C2630	7/6/2015	7/6/2017
02467	Spectrum Analyzer	HP	E7405A	5/10/2016	5/10/2017
P04357	Coaxial Cable	Carolina Wire	RG142	4/4/2016	4/4/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
01647	Bulk Injection Probe	FCC	F-120-1	7/1/2015	7/1/2017
00358	Current Probe	Solar	6741-1	5/31/2016	5/31/2017

Conducted Immunity AC Mains Results		
Mode of operation:	Normal Operation	
Port Tested	**Test Results**	**Test Observations**
AC Mains	No degradation	

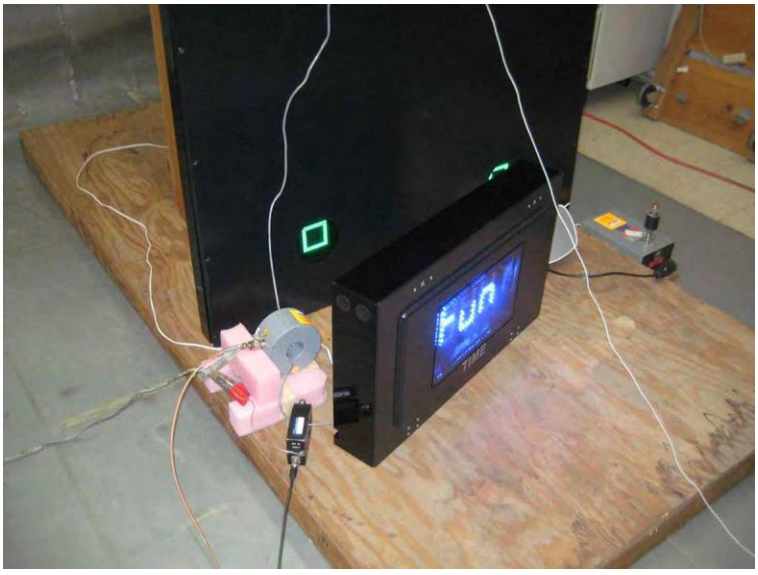
Conducted Immunity IO Lines Results		
Mode of operation:	Normal Operation	
Port Tested	**Test Results**	**Test Observations**
Panel cable	Degradation requiring user intervention	Unexpected increment in score, loss of functionality of the sensors. After the immunity threat is removed the system can be restarted and function normally with a power cycle.
Audio cable	No degradation	

Conducted Immunity Test Summary		
Standard Minimum Performance Criteria:	A	
For this test, the EUT met performance criteria:	MA	Result: Pass

Test Setup Photo(s)



AC



IO

Magnetic Immunity

Test Notes: Immunity threat to simulate mains power frequency radiated magnetic fields.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	NA
Test Method:	EN 61000-4-8	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares that the EUT does not contain devices susceptible to magnetic fields, such as CRT monitors and Hall elements.		
Configuration:	NA		

Voltage Dips and Interrupts

Test Notes: Immunity threat to simulate low-voltage power supply network voltage dips and interruptions of long duration.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	S. Yamamoto
Test Method:	EN 61000-4-11	Test Date(s):	7/11/2016
Test Level:	>95%, 0.5 period 30%, 25 periods >95%, 250 periods		
Declarations:	Modification 1 was in place during testing.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	20	Relative Humidity (%):	40	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
01695	AC Power Supply	Pacific PSC	345AMXT-UPC32	6/2/2016	6/2/2017
01696	AC Power Supply	PPS	AMX-Series Magnetics Module	6/2/2016	6/2/2017

Voltage Dips and Interruptions Results					
Mode of operation:	Normal Operation				
Reduction Ux (%)	**Duration (ms)**	**Minimum Criteria**	**Test Results**	**EUT Meets Criteria**	**Test Observations**
100	10	B	No degradation	A	
30	500	C	No degradation	A	
100	5000	C	Degradation requiring user intervention	C	The equipment turned off then came back on. Operator intervention was needed to set the game back into operational mode.

Voltage Dips and Interruptions Test Summary	
With combination of criteria listed above, the resultant overall assessment is:	Pass

Test Setup Photo(s)



SUPPLEMENTAL INFORMATION

Measurement Uncertainty

Uncertainty Value	Parameter
4.73 dB	Radiated Emissions
3.34 dB	Mains Conducted Emissions
3.30 dB	Disturbance Power

Reported uncertainties represent expanded uncertainties expressed at approximately the 95% confidence level using a coverage factor of k=2.

Emissions Test Details

TESTING PARAMETERS

Unless otherwise indicated, the following configuration parameters are used for equipment setup: The cables were routed consistent with the typical application by varying the configuration of the test sample. Interface cables were connected to the available ports of the test unit. The effect of varying the position of the cables was investigated to find the configuration that produced maximum emissions. Cables were of the type and length specified in the individual requirements. The length of cable that produced maximum emissions was selected.

The equipment under test (EUT) was set up in a manner that represented its normal use, as shown in the setup photographs. Any special conditions required for the EUT to operate normally are identified in the comments that accompany the emissions tables.

The emissions data was taken with a spectrum analyzer or receiver. Incorporating the applicable correction factors for distance, antenna, cable loss and amplifier gain, the data was reduced as shown in the table below. The corrected data was then compared to the applicable emission limits. Preliminary and final measurements were taken in order to ensure that all emissions from the EUT were found and maximized.

CORRECTION FACTORS

The basic spectrum analyzer reading was converted using correction factors as shown in the highest emissions readings in the tables. For radiated emissions in dBµV/m, the spectrum analyzer reading in dBµV was corrected by using the following formula. This reading was then compared to the applicable specification limit. Individual measurements were compared with the displayed limit value in the margin column. The margin was calculated based on subtracting the limit value from the corrected measurement value; a positive margin represents a measurement exceeding the limit, while a negative margin represents a measurement less than the limit.

SAMPLE CALCULATIONS		
	Meter reading	(dBµV)
+	Antenna Factor	(dB/m)
+	Cable Loss	(dB)
-	Distance Correction	(dB)
-	Preamplifier Gain	(dB)
=	Corrected Reading	(dBµV/m)

TEST INSTRUMENTATION AND ANALYZER SETTINGS

The test instrumentation and equipment listed were used to collect the emissions data. A spectrum analyzer or receiver was used for all measurements. Unless otherwise specified, the following table shows the measuring equipment bandwidth settings that were used in designated frequency bands. For testing emissions, an appropriate reference level and a vertical scale size of 10 dB per division were used.

MEASURING EQUIPMENT BANDWIDTH SETTINGS PER FREQUENCY RANGE			
TEST	BEGINNING FREQUENCY	ENDING FREQUENCY	BANDWIDTH SETTING
CONDUCTED EMISSIONS	150 kHz	30 MHz	9 kHz
RADIATED EMISSIONS	9 kHz	150 kHz	200 Hz
RADIATED EMISSIONS	150 kHz	30 MHz	9 kHz
RADIATED EMISSIONS	30 MHz	1000 MHz	120 kHz
RADIATED EMISSIONS	1000 MHz	>1 GHz	1 MHz

SPECTRUM ANALYZER/RECEIVER DETECTOR FUNCTIONS

The notes that accompany the measurements contained in the emissions tables indicate the type of detector function used to obtain the given readings. Unless otherwise noted, all readings were made in the "positive peak" detector mode. Whenever a "quasi-peak" or "average" reading was recorded, the measurement was annotated with a "QP" or an "Ave" on the appropriate rows of the data sheets. In cases where quasi-peak or average limits were employed and data exists for multiple measurement types for the same frequency then the peak measurement was retained in the report for reference, however the numbering for the affected row was removed and an arrow or caret ("^") was placed in the far left-hand column indicating that the row above takes precedence for comparison to the limit. The following paragraphs describe in more detail the detector functions and when they were used to obtain the emissions data.

Peak

In this mode, the spectrum analyzer or receiver recorded all emissions at their peak value as the frequency band selected was scanned. By combining this function with another feature called "peak hold," the measurement device had the ability to measure intermittent or low duty cycle transient emission peak levels. In this mode the measuring device made a slow scan across the frequency band selected and measured the peak emission value found at each frequency across the band.

Quasi-Peak

Quasi-peak measurements were taken using the quasi-peak detector when the true peak values exceeded or were within 2 dB of a quasi-peak specification limit. Additional QP measurements may have been taken at the discretion of the operator.

Average

Average measurements were taken using the average detector when the true peak values exceeded or were within 2 dB of an average specification limit. Additional average measurements may have been taken at the discretion of the operator. If the specification or test procedure requires trace averaging, then the averaging was performed using 100 samples or as required by the specification. All other average measurements are performed using video bandwidth averaging. To make these measurements, the test engineer reduces the video bandwidth on the measuring device until the modulation of the signal is filtered out. At this point the measuring device is set into the linear mode and the scan time is reduced.

Immunity Test Details

Radiated Immunity Field Intensity at 0.4m

Chamber (80-1000)	1V/m Testing	3V/m Testing	10V/m Testing	20V/m Testing	30V/m Testing
Fremont C1	1.5	5.2	11.0	NA	NA
Fremont C2	1.9	2.7	7.1	16.2	NA
Mariposa	1.0	3.3	10.7	NA	NA
Brea	0.6	2.2	12.4	NA	NA
Bothell C1	1.7	3.1	9.0	17.0	25.4
Bothell C2	0.6	1.9	6.8	NA	NA

NA = Not applicable because these levels are not performed in this chamber.